

Lecture notes - social cognition part II

Ivana Konvalinka

November 4, 2025

1 Introduction

As we discussed last week, much of the research within social cognition has focused on individuals responding to computerized "social" stimuli (which are often static), while isolated from an actual interaction with another person. The underlying idea has been that social cognition is a process that goes on within an individual who creates models of other people's mental states and incorporates them with their own. Hence, this approach assumes that the processes that allow us to interact with other people are entirely internalized. This approach has been criticised in that social cognition is fundamentally different when we interact with other people, rather than remain mere observers. In particular, it has been argued that the individual person approach does not tell us how people's perceptions and actions are modulated during real-time interactions with other people. As a result, the joint action approach emerged, with the aim of studying the social mechanisms that underlie engagement in social interaction between two or more people.

Joint action has been defined as "any form of social interaction whereby two or more individuals coordinate their actions in space and time to bring about a change in the environment" [3]. Our everyday lives are filled with joint actions - from coordinating breakfast together with our partner, to walking in step with a colleague, playing a piano duet, or making joint decisions about how to run a company. There are many mechanisms that underlie successful joint actions, depending on the complexity of the joint goal (e.g., walking in step being a very low-level one, while joint decision making requires metacognition and is high-level). Importantly, joint action research allows us to study not only individual mechanisms, but also interpersonal mechanisms - what goes on between two or more people as they interact. If you take a conversation between two people as an example, it would be difficult to make sense of the interaction if the interlocutors were analyzed separately. We may know which person the speech belongs to, but the conversation only makes sense when we analyze it as a whole between two interlocutors.

But designing joint action experiments is very challenging, especially controlling for interactive aspects. People are "noisy" (their actions are uncertain), which means that how interactions develop is difficult to predict and control for. Moreover, many real-time interaction effects are difficult to measure. Think about all the signals and processes that may be exchanged when cooking a meal together with someone. Can you break this down into quantifiable signals and measures?

In order to simplify the scope of joint action research, we will only focus on two types of joint action tasks. The first set of tasks, and probably most researched, investigate how people coordinate their actions in time. The second type focuses on how we make joint decisions together.

2 How we coordinate our actions together

We coordinate our actions together with others often intentionally, to reach a common goal. For example, say we want to move a table together across the room. This requires an understanding that we have the common goal of moving the table to another location; monitoring the other person's task to grab the table at the other end; perhaps producing complementary actions, as you lift the table in front of you to walk forward, while the other lifts the table by resting it on their back to also be able to walk forward; the anticipation of both own and other person's obstacles; and the joint adaptation of forces applied to keep the table from falling. Yet we do this type of task with ease, with little apparent conscious effort, and easily without conversing with the other person.

We also coordinate with others unintentionally. For example, after a concert or opera, when people begin to clap they begin at their own frequencies, but quickly fall into synchrony with one another. Many studies have replicated this effect in the lab, instructing participants to clap or rock (in rocking chairs) at their own preferred frequency, and finding that they still fall into sync, without the intention to do so, even when their natural rhythms are different (e.g., when the natural frequencies of the rocking chairs have been manipulated by putting weights on one of them). People also synchronize other bodily rhythms such as postures, heart rates, and even brain rhythms when they have informational access to each other (i.e., when they can see each other).

Synchronizing with one another has shown to have a role in human social bonding. People are shown to cooperate more, like each other better, and have better rapport after synchronizing their actions together such as walking in step in sync.

So what mechanisms underlie interpersonal coordination or synchronization? We will narrow this down to a few mechanisms.

Studies of joint tapping (e.g., finger tapping together with another person) have shown that **prediction** is a key mechanism facilitating coordination. For example, we infer our partner's next tap (or action) based on their previous tap (or previous action). So if our partner went faster than us on the last tap, we speed up on the next one; or if they went slower, we slow down. Moreover, we **adapt** to our partner's rhythm continuously, and people are shown to coordinate best when they are mutually adaptive (when they both adapt equally to each other, instead of assuming leader-follower roles), given that they have the same task or same level of skill/expertise [2].

When people have different task constraints, such that the task is more difficult for one person, then implicitly (without awareness) negotiating **leader-follower** roles is the better strategy - such that the person with the more difficult task leads, while the one with the easier task follows and takes care of the coordination. Here, leaders and followers are defined as less-adaptive and more-adaptive partners, respectively. For example, the leader focuses more on themselves, setting the tempo - while the follower adapts to the leader, speeding up when the leader speeds up (on the previous tap), and slowing down when the leader has slowed down.

Finally, people **make themselves more predictable** when coordinating with others, particularly on tasks where they do not receive feedback of each other's actions or decisions. For example, on a task where pairs of participants were instructed to press a button at the same time, with no other information of each other or each other's actions, they adopted a strategy where they made the timing of their button presses as predictable as possible (e.g. by speeding up).

There are many other mechanisms at play, such as having a shared goal and a joint commitment to do the joint action in the first place, which are prerequisites for joint action. We also need to jointly attend to the task, etc. We will not go into length with these mechanisms.

3 How we make joint decisions together

One appealing question within joint action research has been, when are two heads better than one? Or when are two people better at accomplishing something together, than the better individual at accomplishing the task alone? This has been particularly relevant to joint decision making. You may have heard the term "wisdom of the crowds"; or alternatively, "too many cooks spoil the broth". So how do these apply to joint decision making?

Wisdom of the crowd refers to the scenario where the opinion of a group of individuals is better than that of a single expert. In 1906, at a county fair in Plymouth, a crowd was asked to estimate the weight of the ox. Surprisingly, the crowd accurately guessed the weight of the ox when their individual guesses were averaged. In other words, their average was closer to the true weight than the estimates of the individual crowd members. Therefore, it was proposed that the aggregation of many independent estimates can outperform the most accurate individual judgement. However, there are some problems with this. This requires independence of individual judgements (otherwise we may conform to a few individuals), and the crowd makes best decisions if made up of diverse opinions.

But social interaction is not about averaging, but about integrating information. To address this, a study by Bahrami and colleagues [1] tested how two people make joint perceptual decisions. Participants in this task, recruited in pairs, viewed brief visual displays individually across two intervals (each lasting only 85 ms). The intervals contained Gabor patches around an imaginary circle. One interval (either the first or second) contained an oddball target with a slightly higher contrast than all

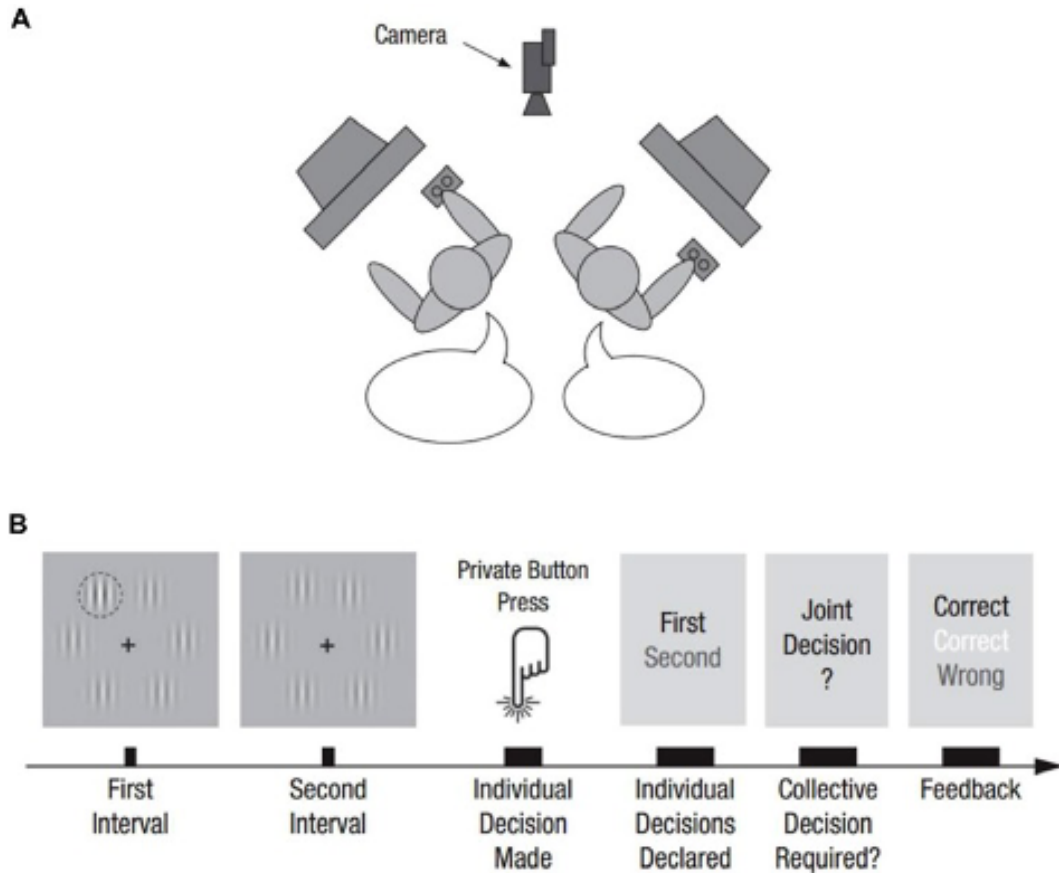


Figure 1: Optimally interacting minds paradigm [1]. A: Pairs of participants engaged in a joint perceptual decision making task. Each made individual decisions, and if they disagreed, they conversed together and had to make a joint decision. B. Experimental paradigm. Each trial consisted of two intervals. In each interval, six Gabor patches were displayed around an imaginary circle for 85 ms. There was an oddball target in one of the intervals that had a higher contrast than the other patches.

the other patches (Figure 1). Following a brief presentation of these visual displays, they were asked to decide individually whether the target appeared in the first or second interval. If the two members of the pair disagreed, they engaged in conversation with each other and had to make a collective decision regarding whether the target appeared in the first or second interval. When the two participants had similar visual sensitivities (measured by fitting psychometric functions to their data), two heads were better than one. But if they had different visual sensitivities, the two heads were worse than the better one. By fitting different models to the data, the authors found that the dyads were Bayes optimal - meaning that they made decisions that maximized their probability of being correct - as long as they communicated their level of confidence on each trial.

We know from work in perception that when we combine information from different senses (e.g., visual and auditory), we have access to both the magnitude of the sensory signals as well as their probability distributions (i.e., means and variances), and that we optimally integrate those together. Here, the authors showed that people also optimally integrate probability distributions across individuals through direct verbal interaction.

References

- [1] B. Bahrami, K. Olsen, P. E. Latham, A. Roepstorff, G. Rees, and C. D. Frith. Optimally interacting minds. *Science*, 329(5995):1081–1085, 2010.

- [2] I. Konvalinka, P. Vuust, A. Roepstorff, and C. D. Frith. Follow you, follow me: continuous mutual prediction and adaptation in joint tapping. *Quarterly journal of experimental psychology*, 63(11):2220–2230, 2010.
- [3] N. Sebanz, H. Bekkering, and G. Knoblich. Joint action: bodies and minds moving together. *Trends in cognitive sciences*, 10(2):70–76, 2006.